

NAVBOT-EN01

communication protocol

Contents

Revision History 3

BLE INFORMATION4

DATA FRAME AGREEMENT4

DATA FRAME DEMONSTRATION 5

Revision History

DOCUMENT HISTORY

[illegible]

BLE INFORMATION

BLE NAME :

navbot_en01-*****

SERVICE UUID:

6E400011-B5A3-F393-E0A9-E50E24DCCA9E

SEND DATA CHARACTERISTIC UUID:

6E400012-B5A3-F393-E0A9-E50E24DCCA9E

RECEIVE DATA CHARACTERISTIC UUID_TX:

6E400013-B5A3-F393-E0A9-E50E24DCCA9E

DATA FRAME AGREEMENT

Byte	Name	Description
Byte 1	Header1	Fixed 0x55
Byte 2	Header2	Fixed 0xAA
Byte 3	Null	Idle
Byte 4	Null	Idle
Byte 5	Null	Idle
Byte 6	roll	Rolling Angle, value range -30 to 30.
Byte 7	height	The height of the robot, with a value range of 32 to 85.
Byte 8	linear_H	Linear speed, with a value range of -200 to 200.
Byte 9	linear_L	
Byte 10	angular	The angular velocity of the robot, with a value range of -100 to 100.
Byte 11	stable	The stable state of the robot: 1 on, 0 off
Byte 12	mode	Fixed 0x01
Byte 13	dir	The movement direction of the robot is defaulted to 0, jump:1
Byte 14	joy_y	The Y-axis data of the joystick, with a value range of -100 to 100.
Byte 15	joy_x	The X-axis data of the joystick, with a value range of -100 to 100.
Byte 16	Null	Idle
Byte 17	Null	Idle
Byte 18	Null	Idle
Byte 19	Null	Idle
Byte 20	Null	Idle

DATA FRAME DEMONSTRATION

55 AA 00 00 00 00 26 0000 00 01 00 00 00 00 00 00 00 00 // height = 0x26(38), stable = 1

After sending this instruction, the robot starts and the leg height is 38mm.

55 AA 00 00 00 00 30 0000 00 01 00 00 00 00 00 00 00 00 // height = 0x30(48), stable = 1 ,

After sending this instruction, the robot starts and the leg height is 48mm.

55 AA 00 00 00 00 30 0000 00 00 00 00 00 00 00 00 00 00 // height = 0x26(38), stable = 0 ,

After sending this instruction, the robot stops and the leg height is 48mm.

55 AA 00 00 00 00 26 0000 00 01 00 00 00 05 00 00 00 00 // height = 0x26(38), stable = 1 ,

joy_y = 0x05(5) ,

After sending this instruction, the robot moves forward at a speed of 5.

55 AA 00 00 00 00 26 0000 00 01 00 00 00 FA 00 00 00 00 // height = 0x26(38), stable = 1 ,

joy_y = 0xFA(-5),

After sending this instruction, the robot moves backward at a speed of 5.

55 AA 00 00 00 00 26 0000 00 01 00 00 00 05 00 00 00 00 // height = 0x26(38), stable = 1 ,

joy_x = 0x05(5) ,

After sending this instruction, the robot rotates clockwise with an angular velocity of 5.

55 AA 00 00 00 00 26 0000 00 01 00 00 00 FA 00 00 00 00 // height = 0x26(38), stable = 1 ,

joy_x = 0xFA(-5),

After sending this instruction, the robot rotates counterclockwise with an angular velocity of 5.

55 AA 00 00 00 00 26 0000 00 01 00 00 01 00 00 00 00 00 // height = 0x26(38), stable = 1 ,

dir = 0x01,

sending this instruction, the robot jumps and is ready.

55 AA 00 00 00 00 26 0000 00 01 00 00 00 00 00 00 00 00 // height = 0x26(38), stable = 1 ,

dir = 0x00,

After sending this instruction, the robot jumps in place.